

Regression 1

Aliaksandr Hubin

February 10 2025

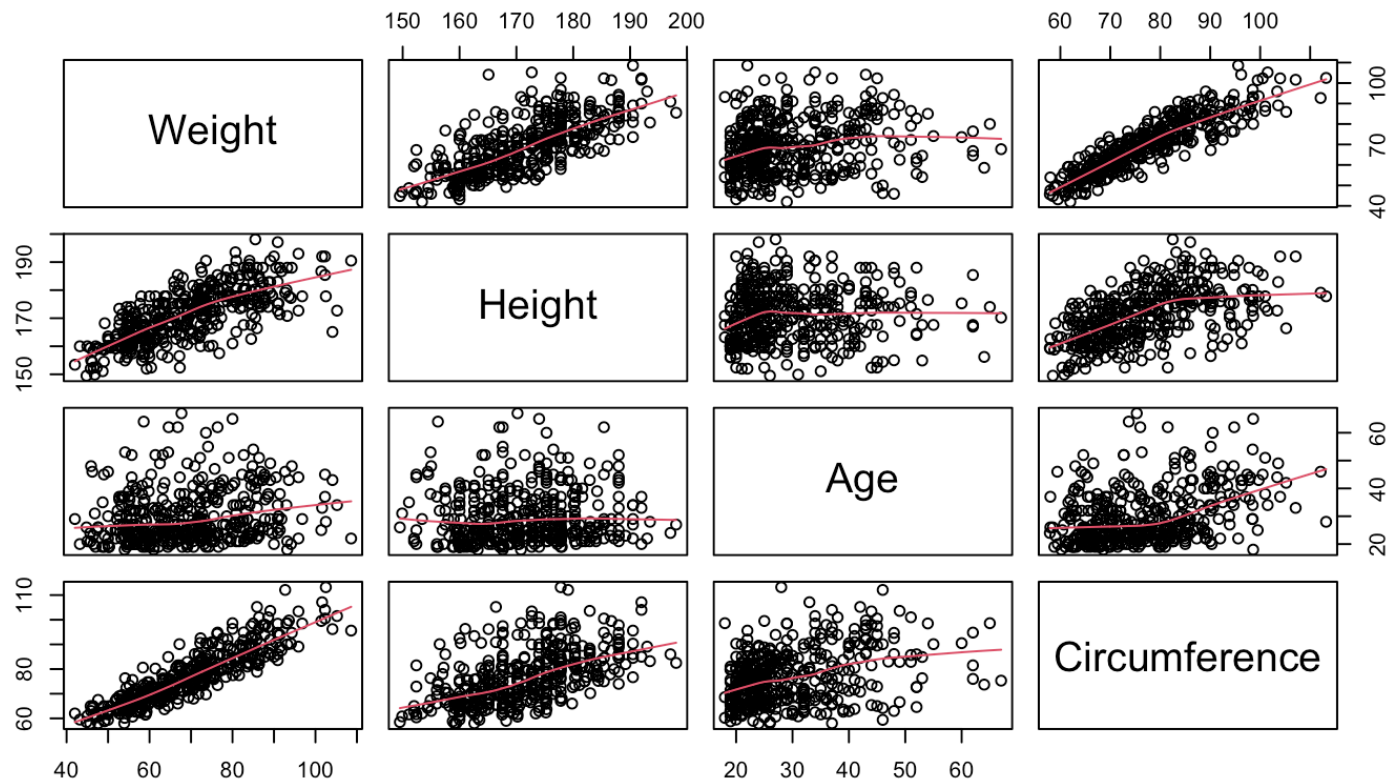
The body data

A motivating example - The body data

```
library(BIAS.data)  
head(bodydata, 10)
```

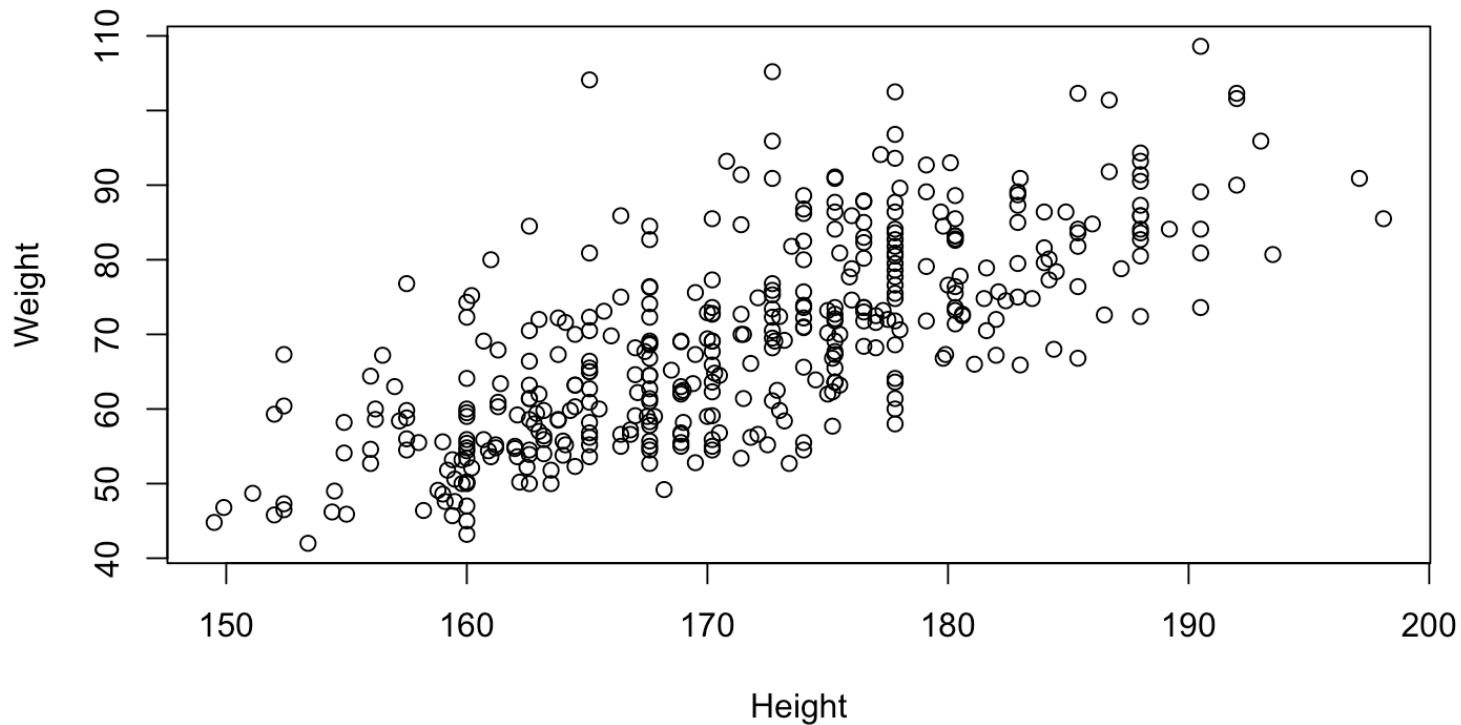
##	Weight	Height	Age	Circumference
## 1	65.6	174.0	21	71.5
## 3	80.7	193.5	28	83.2
## 4	72.6	186.5	23	77.8
## 5	78.8	187.2	22	80.0
## 6	74.8	181.5	21	82.5
## 7	86.4	184.0	26	82.0
## 8	78.4	184.5	27	76.8
## 9	62.0	175.0	23	68.5
## 10	81.6	184.0	21	77.5
## 11	76.6	180.0	23	81.9

Scatterplots



The Weight versus Height

Is there a linear relationship between Weight and Height?



If so, what is the “best” fitting line?

A graphical presentation of a simple linear model

Let's use a web applet to discuss the “best” fitting line:

<http://solve.shinyapps.io/LeastSquaresApp>

The main idea

A continuous response variable, y , is assumed to be linearly dependent on one or several predictor variables, $x_1, x_2, \dots, x_p$

A simple regression model (one predictor)

$$y = \beta_0 + \beta_1 x + \epsilon, \epsilon \sim N(0, \sigma^2)$$

A multiple regression model (multiple predictors)

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \epsilon, \epsilon \sim N(0, \sigma^2)$$

Estimation of regression coefficients

Unknown regression coefficients of a simple regression model are β_0 and β_1

Estimation of parameters is equivalent to finding the “best fitting” model.

Different estimation methods $\Leftrightarrow$ Different definitions of “best”.

- Least squares estimator : The estimates minimize $\sum (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2$
- Equivalent to the maximum likelihood estimator: The estimates make the observed data “most likely”

Maximum likelihood formulation

Recall normal pdf $f(y) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{1}{2} \left(\frac{y-\mu}{\sigma}\right)^2\right\}$

Derivations for maxima

Estimation of unexplained variance

As soon as we have defined the “best” fitting model, we have also defined all distances between the observations and the estimated model.

These distances are the residuals:

$$d_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)$$

for $i = 1, \dots, n$

An unbiased estimator for the error variance σ^2 depends on these distances through

$$\hat{\sigma}^2 = \sum d_i^2 / (n - p - 1) = \frac{SSE}{(n - p - 1)} = MSE$$

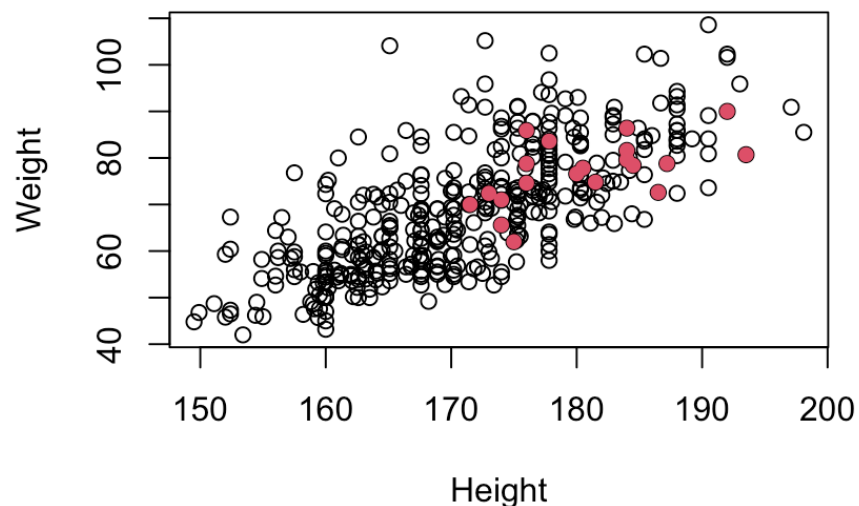
where p is the number of predictors.

Example: Regression in R (Model 1)

We use samples 1 to 20 of the bodydata as example data (training set)

$$y_i = \beta_0 + \beta_1 x_{1i} + \epsilon_i$$

where y is Weight, x_1 is Height and $i = 1, \dots, 20$.



Example of model fit in R Studio

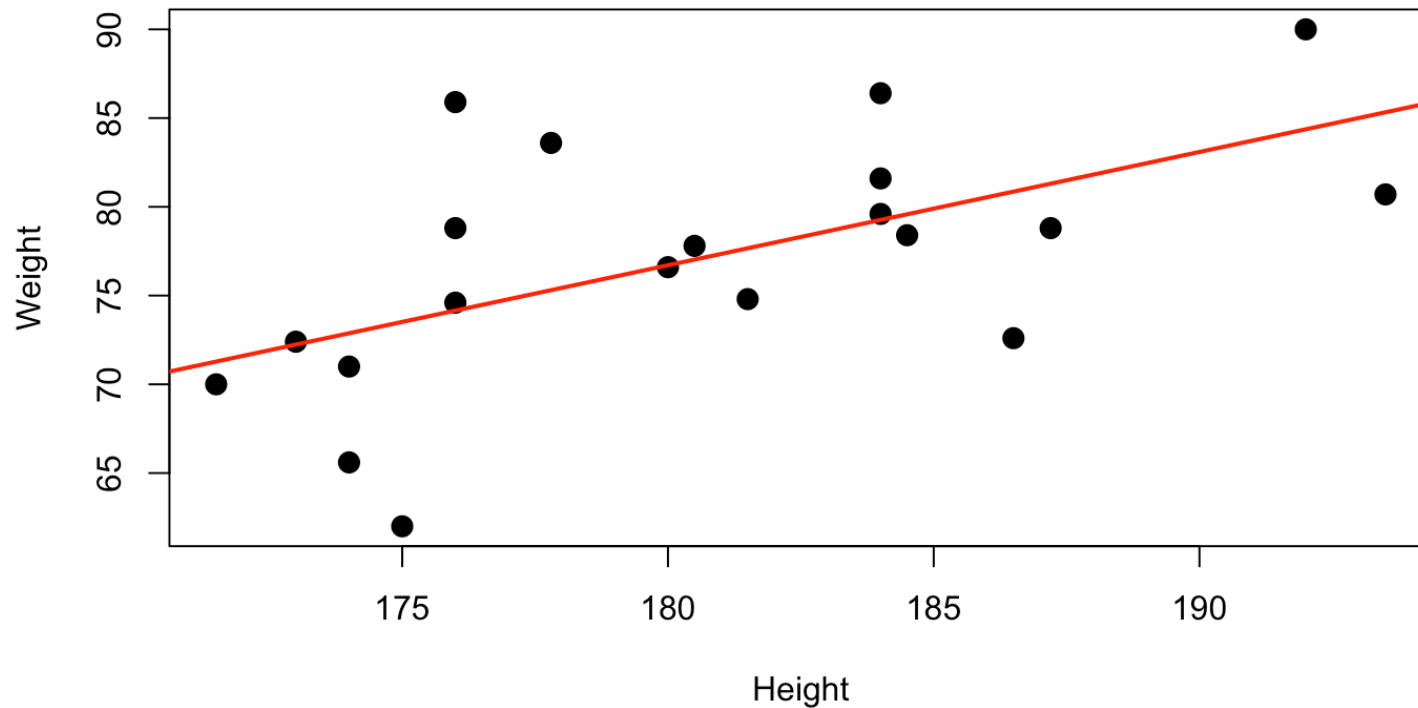
```
LinearModel.1 <- lm(Weight ~ Height, data=bodydata, subset=1:20)  
summary(LinearModel.1)
```

Output

```
##
## Call:
## lm(formula = Weight ~ Height, data = bodydata, subset = 1:20)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.5160  -2.5964   0.0261   2.9141  11.7454
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -38.2310     38.3605  -0.997  0.33216
## Height        0.6386      0.2123   3.007  0.00757 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.839 on 18 degrees of freedom
## Multiple R-squared:  0.3344, Adjusted R-squared:  0.2974
## F-statistic: 9.043 on 1 and 18 DF,  p-value: 0.007566
```

The fitted line plot

```
plot(Weight~Height, data=bodydata, subset=1:20, pch=20, cex=2)  
abline(LinearModel.1, lwd=2, col="red")
```

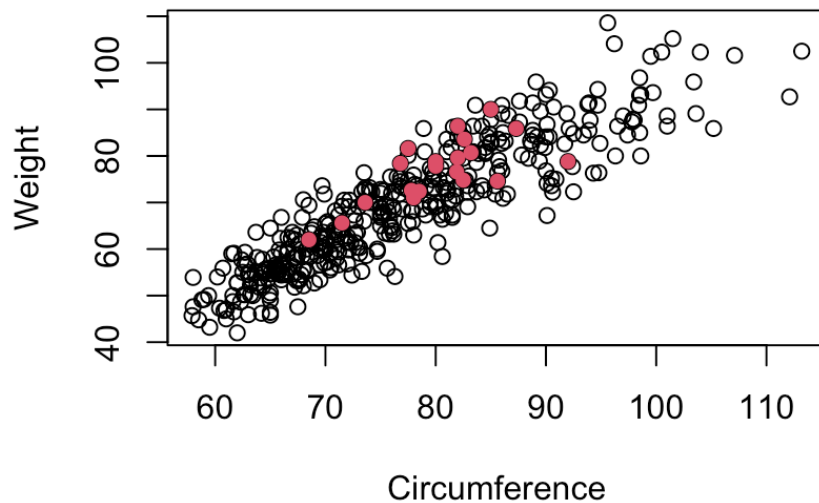


Weight versus Circumference (Model 2)

Now let:

$$y_i = \beta_0 + \beta_1 x_{2i} + \epsilon_i$$

where y still is Weight, x_2 is Circumference and $i = 1, \dots, 20$.



R Commander output

```
##
## Call:
## lm(formula = Weight ~ Circumference, data = bodydata, subset = 1:20)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.937 -3.540  0.038  2.956  8.659
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.6737    17.1036   0.215 0.832343
## Circumference    0.9137     0.2125   4.300 0.000431 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.027 on 18 degrees of freedom
## Multiple R-squared:  0.5067, Adjusted R-squared:  0.4793
## F-statistic: 18.49 on 1 and 18 DF,  p-value: 0.0004311
```

How can we claim that this model gives a better fit than model1?

A measure of goodness of fit, the R^2

The unconditional (sample) variance of Weight is:

$$\sum (y_i - \bar{y})^2 / (n - 1) = \frac{SST}{(n - 1)}$$

Hence, SST (in some literature SSY) is a measure of total variability in y , the response

Further, SSE is a measure of unexplained variability in y

So, a measure of explained variability is

$$R^2 = \frac{SST - SSE}{SST} = 1 - \frac{SSE}{SST}$$

Multiple linear regression (Model 3)

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \epsilon_i$$

where y is Weight, x_1 is Height x_2 is Circumference and $i = 1, \dots, 20$.

R Studio output (Model 3)

```
##  
## Call:  
## lm(formula = Weight ~ Height + Circumference, data = bodydata,  
##      subset = 1:20)  
##  
## Residuals:  
##      Min      1Q  Median      3Q      Max   
## -5.3189 -3.5363 -0.7817  2.8028  6.3974   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  -70.8202    28.4518  -2.489  0.023464 *      
## Height         0.4731     0.1566   3.021  0.007698 **     
## Circumference  0.7777     0.1820   4.273  0.000515 ***    
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 4.172 on 17 degrees of freedom  
## Multiple R-squared:  0.679, Adjusted R-squared:  0.6413   
## F-statistic: 17.98 on 2 and 17 DF,  p-value: 6.38e-05
```

Three predictors...(Model 4)

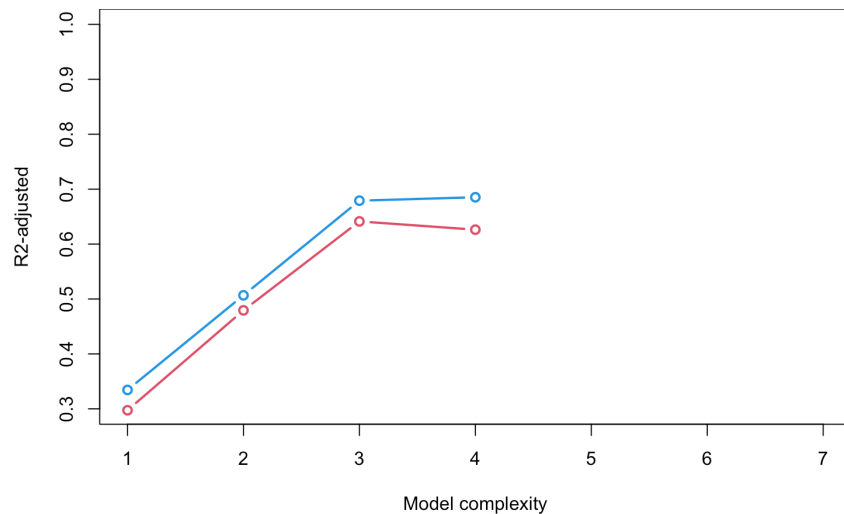
```
##  
## Call:  
## lm(formula = Weight ~ Height + Circumference + Age, data = bodydata,  
##      subset = 1:20)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -5.4088 -3.3900 -0.9123  3.0403  6.9827   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  -66.5785    30.0098  -2.219  0.041330 *     
## Height         0.4768     0.1600   2.981  0.008831 **    
## Circumference  0.7769     0.1858   4.181  0.000706 ***   
## Age          -0.2094     0.3731  -0.561  0.582352      
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 4.259 on 16 degrees of freedom  
## Multiple R-squared:  0.6852, Adjusted R-squared:  0.6262   
## F-statistic: 11.61 on 3 and 16 DF,  p-value: 0.0002729
```

R^2 -adjusted

$R^2 = 1 - \frac{SSE}{SST}$, but the Problem: R^2 always increases when adding pred

$$R^2_{\text{adj}} = 1 - \frac{\hat{\sigma}^2}{\widehat{Var}(y)} = 1 - \frac{SSE/(n - p - 1)}{SST/(n - 1)}$$

The Solution: R^2_{adjusted} penalizes adding parameters



Hypothesis Test for Regression Coefficients

We may test the **conditional significance** of any predictor using a t-test

Hypotheses:

- $H_0 : \beta_j = 0$ (predictor has **no effect** on response)
- $H_1 : \beta_j \neq 0$ (predictor **does have an effect**)

Test statistic:

$$T_j = \frac{\hat{\beta}_j - 0}{SE(\hat{\beta}_j)} \sim t_{n-p-1}$$

Visual explanation

Decision rule

The **p-value** is the probability of observing a test statistic as extreme or more extreme than the observed value, **assuming** H_0 is true

$$\text{p-value} = P(|T| \geq |T_{\text{obs}}|) = 2 \times P(T > |T_{\text{obs}}|)$$

- **p-value is small** (e.g., 0.002): It's **very unlikely** to observe such an extreme
- T if H_0 were true $\rightarrow$ Strong evidence against $H_0 \rightarrow$ **Reject H_0**

At significance level of 0.05:

- Reject H_0 if $|T_j| > \tau_{\alpha/2, n-p-1}$ (i.e., p-value < 0.05)
- Fail to reject H_0 if $|T_j| \leq \tau_{\alpha/2, n-p-1}$ (i.e., p-value ≥ 0.05)

Example: Model 4 Significance tests

##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	-66.5785315	30.0098013	-2.2185596	0.0413297267
## Height	0.4768387	0.1599838	2.9805444	0.0088305547
## Circumference	0.7768555	0.1858173	4.1807501	0.0007063713
## Age	-0.2094104	0.3730630	-0.5613273	0.5823518801

Key insight:

- Age is not significant ($p = 0.582 \gg 0.05$)
- Explains why adding Age slightly reduces R^2_{adj} - Model 3 (without Age) is **better choice** than Model 4
- Other predictors are significant

Confidence interval for regression coefficients

A $(1 - \alpha) \cdot 100\%$ for β_j is mathematically given by:

$$\left[\hat{\beta}_j \pm \tau_{\alpha/2, (n-p-1)} SE(\hat{\beta}_j) \right]$$

In RStudio we can get these (ex. for model 4) by

```
confint(LinearModel.4, level=0.95)
```

##		2.5 %	97.5 %
## (Intercept)	-130.1964683	-2.9605946	
## Height	0.1376883	0.8159891	
## Circumference	0.3829405	1.1707706	
## Age	-1.0002686	0.5814477	

Prediction

A fitted model can be used for prediction of new observations

According to model 4, what is the predicted Weight of some person with these values?

```
##   Height Circumference Age
## 1    171           76  30
```

$$\begin{aligned}\hat{y} &= -66.579 + 0.4768 \cdot 171 + 0.7769 \cdot 76 - 0.2094 \cdot 30 \\ &= 67.72\end{aligned}$$

```
newobs <- data.frame('Height'=171, 'Circumference'=76, 'Age'=30)
predict.lm(LinearModel.4, newdata=newobs)
```

```
##      1
## 67.7196
```

Other intervals in Prediction output

Confidence interval for $E(y|x_1, \dots, x_p)$

```
predict.lm(LinearModel.4, newdata=newobs,  
           level=0.95, interval=c("confidence"))
```

```
##          fit      lwr      upr  
## 1 67.7196 60.98252 74.45668
```

Prediction interval for $y|x_1, \dots, x_p$

```
predict.lm(LinearModel.4, newdata=newobs,  
           level=0.95, interval=c("prediction"))
```

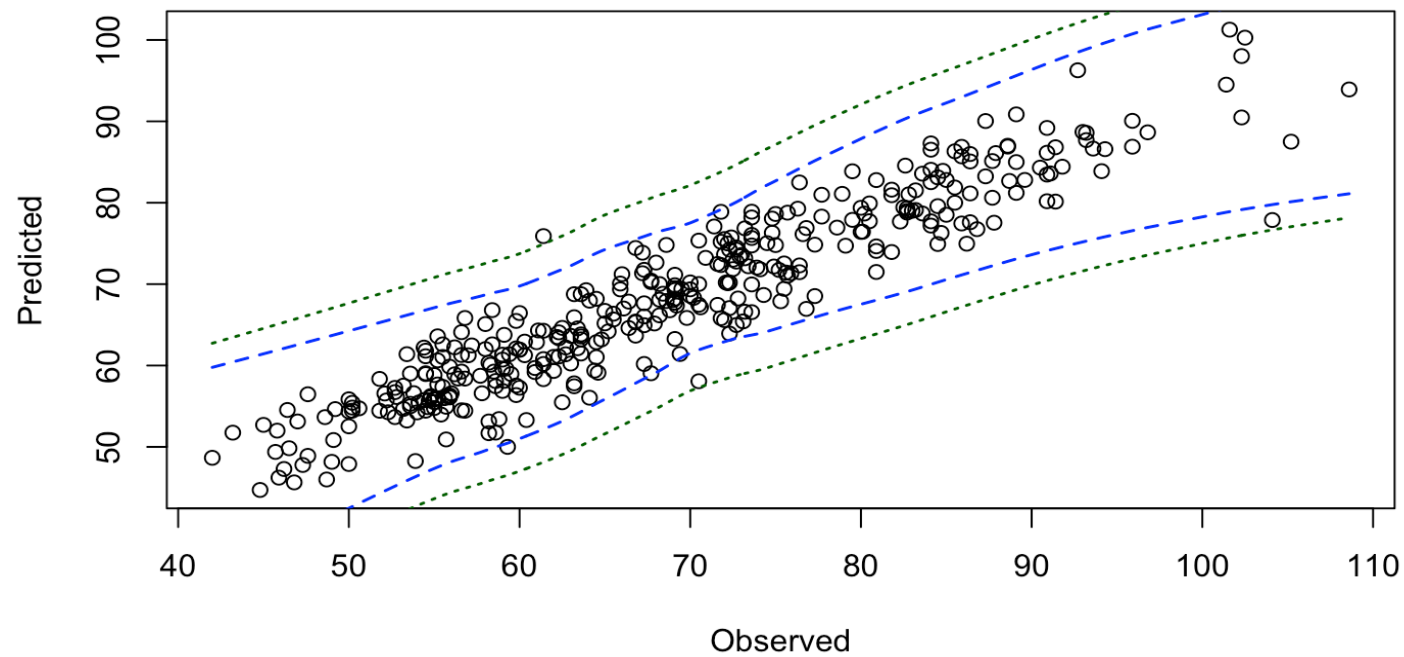
```
##          fit      lwr      upr  
## 1 67.7196 56.45463 78.98456
```

Compare real and predicted values

Let's predict the 387 extra observations and compare with actual observations!

##		fit	lwr	upr
##	25	73.94436	64.06824	83.82047
##	26	82.80148	72.72603	92.87693
##	27	81.03059	71.44393	90.61725
##	28	71.46852	61.81893	81.11811
##	29	65.93065	55.27452	76.58677
##	31	76.31124	66.90951	85.71296
##	32	69.35871	59.61763	79.09979
##	33	75.73802	65.67258	85.80346
##	34	82.52507	72.84459	92.20555
##	35	67.71321	57.86212	77.56431
##	37	73.86325	64.29154	83.43496
##	39	74.81258	65.46899	84.15617

Predicted vs observed - plot



Checking the prediction intervals

How many observations fell within the prediction intervals?

```
largerthanlower <- (newWeight > pred[,2])  
smallerthanupper <- (newWeight < pred[,3])  
inInterval <- which(largerthanlower & smallerthanupper)  
print(length(inInterval))
```

```
## [1] 381
```

Proportion of observations within prediction intervals:

```
length(inInterval)/length(newWeight)
```

```
## [1] 0.9844961
```

What should we expect?